2026 Greenbrier Science Olympiad Invitational

January 10, 2026

Disease Detectives B/C

ANSWER KEY

For ease of grading, point values are listed beside each answer.

Part 1 (give credit for definitions that approximate what is listed here)

1. A disease that is always present in a population at a given baseline level — **1**
2. An outbreak of disease affecting animals — **1**
3. A disease that is passed between animals and humans — **1**
4. An infection acquired in a hospital or other healthcare setting (also known as a "hospital-acquired infection" or "HAI") — **1**
5. The severity of disease caused by an infectious agent — **1**
6. An inanimate object that can transmit disease — **1**
7. A disease outbreak affecting a wide geographic area (spans multiple countries, can be worldwide) — **1**
8. The ability of a statistical test to return true negatives — **1**
9. The passing of a disease from a mother to her child (transmission can happen pre-birth, during birth, or post-birth) — **1**
10. On average, the number of individuals in a wholly susceptible population who will be infected by one infected person — **1**
11. Human immunodeficiency virus — **1**
12. Centers for Disease Control and Prevention *(accept "Centers for Disease Control")* — **1**
13. Médecins Sans Frontières *(accept "Doctors without Borders")* — **1**
14. Severe acute respiratory syndrome — **1**
15. Shiga toxin-producing *E. coli* (or "Shigatoxigenic *E. coli*") — **1**
16. Health Insurance Portability and Accountability Act — **1**
17. Coronavirus disease — **1**
18. Morbidity and Mortality Weekly Report — **1**
19. Tetanus, diphtheria, and pertussis — **1**
20. Measles, mumps, and rubella — **1**
21. D, B, C, F, A, G, E — **3 (must all be in correct order, no partial credit)**
    *Justinian plague **541-49**; Black death **1346-53**; Broad St. cholera **1854**; Spanish flu **1918-20**; Swine flu **2009-10**; Flint water **2014-19**; COVID **2019-***
22. Chain of infection — **1**
23. Fomite — **1**
24. Pathogenicity, virulence — **1 each**
25. Chain of infection — **1**
26. Incubation period, latent period — **1 each**
27. GRADE — **1**
28. Immunization, inoculation, vaccination — **1 each**
29. Syndromic, index case — **1 each**
30. Efficacy, efficiency — **1 each**
31. CDC — **1**
32. Carrier, vector — **1 each**
33. Endemic, hyperendemic, holoendemic — **1 each**
34. Vaccination — **1**

Part 2

35. **S** (parasite), caused by genus *Plasmodium* — **1**
36. **S** (parasite), caused by genus *Trichinella* — **1**
37. **B** (bacterium), caused by *Yersinia pestis* — **1**

38. **R** (prion) — **1**
39. **V** (virus), caused by SARS-CoV-2 — **1**
40. **S** (parasite), caused by genus *Cryptosporidium* — **1**
41. **V** (virus), caused by Herpes simplex virus (HSV-1 and HSV-2) — **1**
42. **R** (prion) — **1**
43. **B** (bacterium), caused by genus *Salmonella* — **1**
44. **B** (bacterium), caused by *Corynebacterium diphtheriae* — **1**
45. **B** (bacterium), caused by genus *Legionella* — **1**
46. **F** (fungus), caused by *Histoplasma capsulatum* — **1**
47. **F** (fungus), caused by *Pneumocystis jirovecii* — **1**
48. **B** (bacterium), caused by *Bordetella pertussis* — **1**
49. **B** (bacterium), caused by *Bacillus anthracis* — **1**
50. Tertiary — **1**
51. Primary — **1**
52. Secondary — **1**
53. Primary — **1**
54. Quaternary — **1**
55. Secondary — **1**
56. Secondary — **1**

Part 3 (give credit for answers that approximate what is listed here)

57. The double-blind design eliminates potential bias on the part of the researcher and the participant. — **2**
58. Case-control, since participants are split into "Cases" and "Controls" from the start, meaning cases will not have to be found over the course of the study. Cohort studies are poorly suited for rare diseases because they group by exposure and it is fairly unlikely to get an adequate sample size of cases for a rare disease. — **3**
59. Herd immunity protects individuals who are unable to get vaccinated due to comorbidities or other conditions which leave them more immunocompromised. Failure to achieve herd immunity is a risk because it puts these individuals in danger of infection when they cannot protect themselves as well. — **3**
60. Data collection, data analysis, data interpretation, data dissemination, link to action — **2**
61. Propagated; messier shape reflects irregularity of person-to-person transmission and lack of single common source. — **2**
62. None; a case must have laboratory testing before it is confirmed. — **2**
63. C → E → S — **1**
64. There is no causal path between I and J — **1**
65. The following are all acceptable answers: — **1**
    C → G → D → Z
    C → E → D → Z
    C → E → M → D → Z
66. There is no causal path between J and E — **1**
67. There are two possible colliders: D, E, S — **1**
68. There are four possible intermediaries: D, E, G, M — **1**
69. A and D (must get answer exactly right to receive points; no partial credit) — **2**
70. Epidemic curve — **1**
71. Continuous common source (or "continuous source") — **1**
72. Two years — **1**
73. Week 43, 2021 — **1**
74. Accept any answer between Week 47, 2022 and Week 3, 2023 — **1**
75. Maximum is 46 cases; minimum is 9 cases — **1 each, 2 total**

76. Award half a point each for agent (meningitis), location (Michigan), time frame (2021–23). Award full two points for mentioning all three of the above. — **0.5 each, up to 2 total**
77. Relative risk — **1**
78. [74 / (74 + 29)] / [31 / (31 + 103)] = 9916 / 3193 = 3.11 — **2**
79. People who were exposed had 3.11 times the risk of disease as compared to people who were unexposed. — **2**
80. Yes, the magnitude of the order of association indicates that the exposure was responsible. — **3**
81. 245 with disease, 432 total study population; 245 / 432 = 0.57 — **2**
82. 173 exposed with disease; 88 exposed without disease; 173 / 261 = 0.66 — **1**
83. No, since the study was cross-sectional (exposure and disease measured at the same time), we cannot draw any conclusions about causality. — **3**
84. Odds ratio — **1**
85. Exposure A: (116*[90+308]) / (162*[124+17]) = 2.02 — **2**
    Exposure B: (124*[162+308]) / [90*[116+17]) = 4.87 — **2 (up to 4 total for the whole question)**
    Recall that, when calculating OR for exposure A, the "non-exposed" group includes "Exposure B" and "No exposure" since neither group received Exposure A!
86. Exposure B, based on the magnitude of the measure of association (4.87 is over twice the association as 2.02).

Greenbrier Invitational - Repaired practice edition - Answer key

Division C | Disease Detectives | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

33. (6 points) Authored replacement for source item Question 33: Solve this case and explain your reasoning: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

reasoning: Selection is based on banquet attendance, not illness status.
Exposure and outcome are reconstructed for the full defined cohort.
Food-specific risks and RR can be estimated if ascertainment and denominators are adequate. (3 points).

conclusion: This is a retrospective cohort, despite collecting the information after the event. (3 points).

54. (2 points) Authored replacement for source item Question 54: Define “Incubation period” in the context of Disease Detectives.

answer: Time from infection to symptom onset. (2 points).

55. (4 points) Authored replacement for source item Question 55: Distinguish infection and reporting dates.

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

answer: Infection is acquisition of the agent; reporting occurs after detection and administrative processes and can be substantially later. (4 points).

69. (2 points) Authored replacement for source item Question 69: Which measure concerns new cases?

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

A. Reporting delay

B. Incidence

C. Point prevalence

D. Mean disease duration

Correct: B